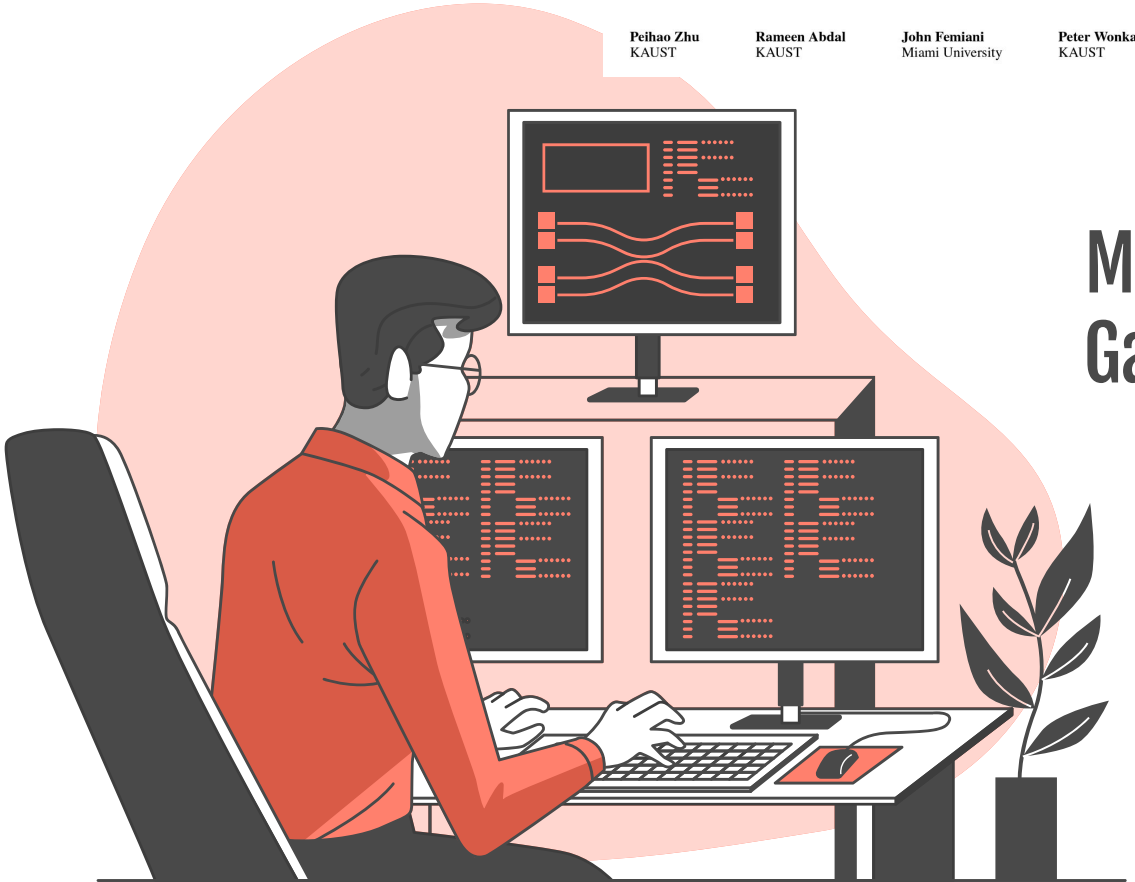
Published as a conference paper at ICLR 2022

MIND THE GAP: DOMAIN GAP CONTROL FOR SINGLE SHOT DOMAIN ADAPTATION FOR GENERATIVE ADVERSARIAL NETWORKS

Peihao Zhu KAUST
Rameen Abdal KAUST
John Femiani Miami University
Peter Wonka KAUST



Mind the Gap: Domain Gap Control for Single Shot Domain Adaptation for GANs

- Dheemant Jallepalli (B20BB015)

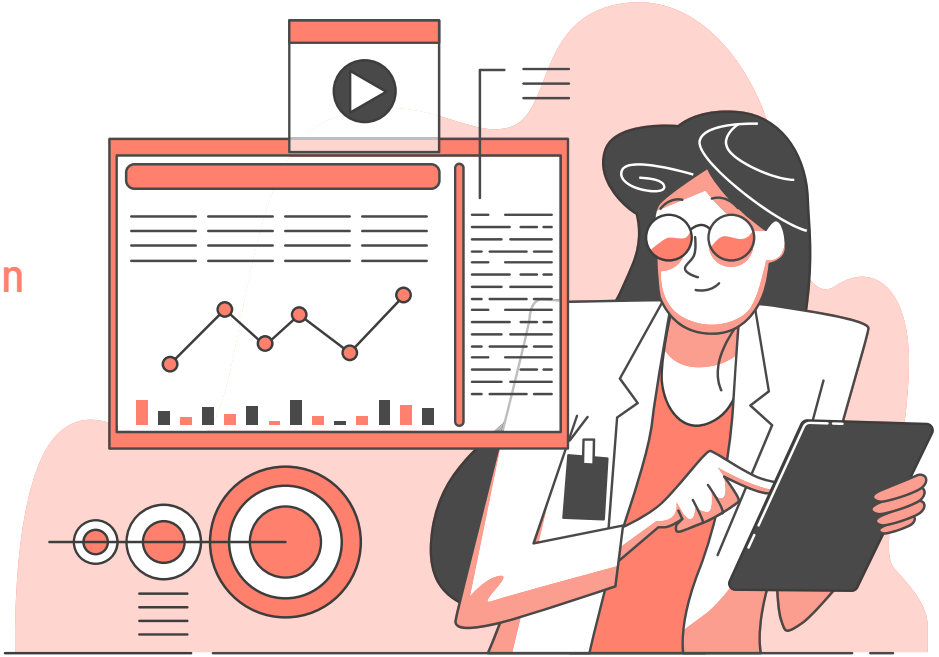
Table of Contents

01 **Introduction**
Briefing!

02 **Implementation**
MTG playground

03 **Change in Implementation**
Retrieve InStyle playground

04 **ChatGPT Question**
Ask! Ask! Verify!!



Introduction

The **proposed method** for **one-shot domain adaptation** utilizes a **pre-trained StyleGAN** generator and a **pre-trained CLIP** model to translate any output of the **trained GAN** from **domain A** to **domain B** using a single **reference** image from **domain B**.
The **two GANs share** a **latent space**, allowing a single latent code to generate two corresponding images, one in domain A and one in domain B.

The proposed method achieves higher visual quality than the **state of the art** and allows for improved **control over** the **domain gap**. The method is realized through the use of **several new regularizers** that **prevent overfitting** and **preserve** the **identities** of the source domain images.

Providing **more freedom** to **control** the **similarity** between images across domains.
The results demonstrate significant visual improvements and multiple applications.
Components leveraged, are **StyleGAN2, CLIP, StyleGAN-NADA, and II2S.**

Implementation

Aligned face

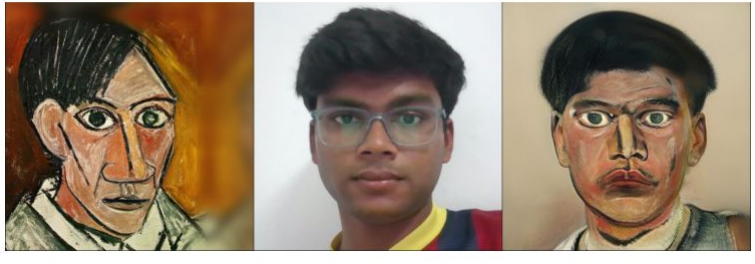


Embedding result



Embedding an input image into the latent space of a pre-trained StyleGAN2 generator, which allows the image to be manipulated and synthesized in the **StyleGAN2** style.
II2S is an embedding method that stands for "**In-Domain Image-to-StyleGAN Inversion**", which is a technique for inverting an image into the latent space of a pre-trained StyleGAN2 generator by minimizing the distance between the generated image and the input image. The II2S method typically results in higher quality embeddings compared to other methods, but takes longer to compute.

Transferred input



Generating image results by transferring the style of a reference image onto a given input face image using a pre-trained StyleGAN2 generator. Where firs, it generates random latent vectors and applies style mixing to create multiple styles for each latent vector. Interpolates between the given style latent vector and the style latent vectors of the generated samples by a specified color interpolation factor

Transferred random samples



Implementation and Results













Multimodal Texture Generation (MTG)

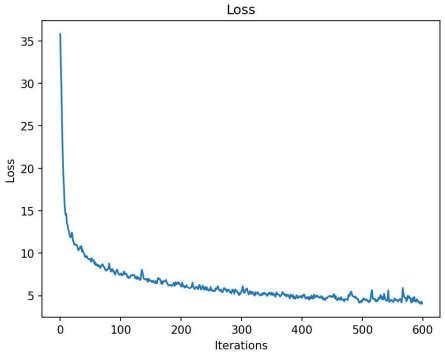
We upload an image for example I am using my photo, which will be used as a style image. The style image will be embedded, and the latents of the input and sample faces will be modified to resemble the style image. Then we train a generator that can generate images that combine the style of the input face and the sample face.

The training is done by **minimizing a weighted sum of several losses**, including **L2 loss**, **LPIPS loss**, that ensures the generated image is semantically similar to the input image when projected through a **CLIP** model. The training is performed for a specified number of iterations, and the resulting generator checkpoints are saved every specified number of iterations.

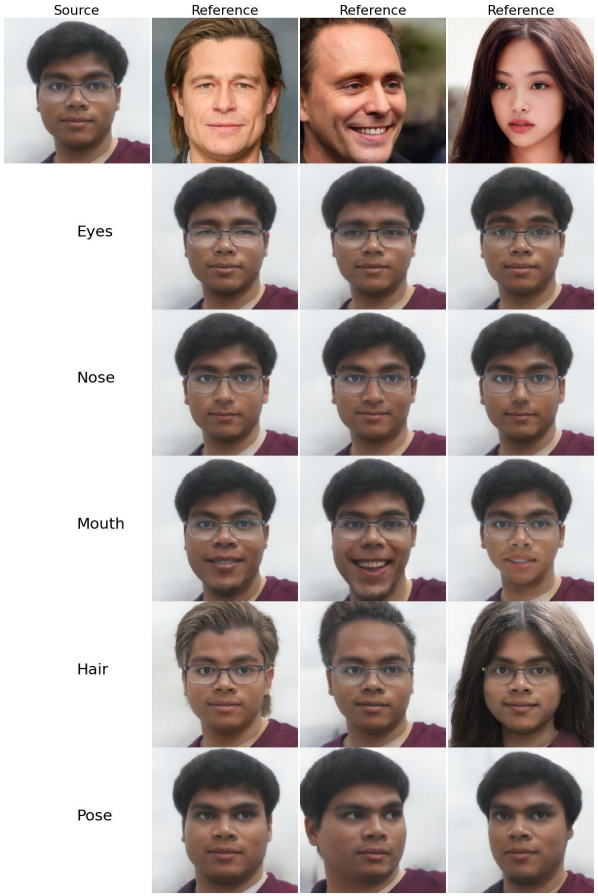
Final result:



Loss after each iter:



New Application using RIS



Final result

Retrieve In Style (RIS)

The main purpose of RIS was to segmentation for the classes we are interested in and correspond the kmeans clusters to those classes. And interpolating into the reference image from ffhq dataset itself.

The earlier implementation used II2S here we will use the e4e method which is an embedding method that stands for "encoding from human to emotion", which uses a pretrained VGG-based encoder to map an input image to a latent code, which is then fed to a generator network to synthesize a new image, as it is faster to compute.

The generator is now used to interpolate between the features of two images, such as eyes, hair, nose, mouth, and pose, and displays the resulting images. The interpolation is done by blending the input source image with a reference image, and gradually changing the weight of specific features.

As we can see we we have changed specific styles in the image depending on the different segmentation labels that were available



Figure shows image segmented based on different attributes in the face, i will be using my image here.



CHAT GPT



https://github.com/ZPdesu/MindTheGap.git

Use the github repo and can you repupose it for new applications

The GitHub repo you linked to contains the implementation of MindTheGap (MTG) GAN, which is a method for style transfer in images.

To repurpose it for a new application, you would need to modify the existing code or create new code that takes advantage of the underlying architecture of the MTG GAN model. Here are a few potential ideas:

1. Video Style Transfer: Modify the existing code to work with video files instead of images. This could be used to create a video that changes its style over time, or to transfer the style of one video onto another.
2. Artistic Text Generation: Use the MTG GAN as a starting point for generating artistic text. Instead of using images as input, use text descriptions of a desired style or mood to generate a unique artistic font or calligraphy style.
3. Music Style Transfer: Apply the MTG GAN model to music files to transfer the style of one musician onto another. This could be used to create new mashup tracks or to apply the style of a classical composer onto modern pop music.
4. Interactive Style Transfer: Build an interactive application that allows users to change the style of an image in real-time using a custom interface. This could be used in photo editing software or as part of an augmented reality app.

Of course, these are just a few potential ideas, and the possibilities are only limited by your imagination and technical expertise. Repurposing existing code can be a great starting point for exploring new applications and pushing the boundaries of what is possible with machine learning.

Regenerate response

Send a message.

Free Research Preview. ChatGPT may produce inaccurate information about people, places, or facts. ChatGPT May 3 Version

Though ChatGPT was able to understand the code and write the whole context and summaries on it, it couldn't code properly and due to its limited knowledge of the world , it has multiple restrictions.

Well, for now I think it actually wanted to code for the following as shown as a new application.

But, the codes were quite improper and they weren't actually accessing the repository and its contents.

Thanks!

Do you have any questions?

jallepalli.1@iitj.ac.in

IIT Jodhpur

Published as a conference paper at ICLR 2022

MIND THE GAP: DOMAIN GAP CONTROL FOR SINGLE SHOT DOMAIN ADAPTATION FOR GENERATIVE ADVERSARIAL NETWORKS

Peihao Zhu KAUST
Rameen Abdal KAUST
John Femiani Miami University
Peter Wonka KAUST


slidesgo